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The Orchestrated Depot: Closing the gap between planning and reality in electric bus operations

A practical blueprint for operators to synchronise scheduling, charging, and depot execution using open standards and a digital control tower.

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Executive Summary

05:47. The temperature has dropped overnight. Twelve buses are due out at 06:30. One charger is offline and two vehicles are charging slower than planned. In a diesel fleet this is an inconvenience. In an electric fleet it is a service failure waiting to happen.

Electrification turns energy into a hard operational constraint. State of Charge is no longer a nice-to-know metric; it is the boundary condition for every duty, swap, and recovery action. Yet most operators still run three separate worlds: scheduling, charging control, and real-time operations.

The result is latency. Problems that could be solved at 22:00 are only discovered at 05:30 when options are limited, staff are scarce, and the first departures are minutes away.

The orchestrated depot is a two-layer approach: keep specialist systems in place, then add a unifying orchestration layer (a digital control tower) that builds a single operational picture, detects conflicts early, and proposes executable actions.

Operators that introduce orchestration typically see materially higher pull-out reliability, 15-25% energy cost reduction through smart charging, less manual coordination effort, and fewer stranded-vehicle incidents. In many cases, energy savings alone can cover the first year of investment.

Why platform, why now: electric operations have become a real-time, multi-constraint system (timetable, energy, vehicle capability, depot capacity, maintenance windows). When those constraints are split across planning tools, charging systems and control-room workflows, decision latency is inevitable. Orchestration treats this as a platform problem: a governed operational backbone that unifies data, exposes events and executes workflows so optimisation and AI can be added without rebuilding integrations or systems each time.

What is the Orchestrated Depot?

The Orchestrated Depot is an integration and decision layer that sits above existing scheduling, charging, and telematics systems. It does not replace them. It connects them, normalises their data, and continuously reconciles plan versus reality so the operation can intervene early and confidently.

Think of it as the execution brain between: (1) the plan (blocks, duties, allocations), (2) the energy system (chargers and grid constraints), and (3) the live fleet (position and charge state).

Core capabilities

Capability	What It Does	Who Uses It
Real-Time SoC Visibility	Tracks every vehicle's actual charge state against its planned trajectory, highlighting deviations as they emerge	Dispatchers, Depot Supervisors
Predictive Pull-Out Alerts	Flags at-risk vehicles hours before departure, not minutes	Operations Managers

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Smart Charging Orchestration	Optimises charging to Time-of-Use/spot market rates and grid capacity constraints	Energy Managers
Automated Conflict Resolution	Proposes vehicle swaps filtered by charge, route, and crew compatibility—across multiple depots	Dispatchers
Forward SoC Trajectory	Predicts end-of-duty and intermediate SoC including opportunity charging events	Control Room, Dispatchers

Operating models: one platform, different challenges

Electric bus operations generally fall into two paradigms. Both benefit from orchestration, but the dominant failure modes and optimisation targets differ.

Model	Characteristics	Primary Orchestration Challenge
Depot-centric	Buses depart at 100% SoC, return depleted, charge overnight	Guaranteeing morning pull-out; optimising overnight charging windows
Opportunity charging	Multiple charging moments per day via pantographs or fast chargers at termini and depots	Real-time SoC trajectory management; same-day cascade prevention

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1 The Operational Reality: Why Electric Fleets Are Different

A diesel bus is operationally autonomous: refuel in minutes and return to service. An electric bus is energy-constrained: charging is time-bound, capacity-bound, and coupled to the duty plan. This coupling creates new operational failure modes.

1.1 The numbers that matter

Range and charging performance are not constants. They vary by weather, HVAC demand, traffic, and infrastructure constraints. Orchestration exists to make those variabilities manageable.

Factor	Impact	Source
Cold weather (-4°C to 0°C)	48% increase in energy consumption; 33-39% range reduction	Gu et al., Cornell/TCAT, Transportation Research Part D (2025); CAA Canada Winter EV Study (2025)
Heating demand (extreme cold)	50%+ of total energy consumption	UITP Bus Committee, Thermal Comfort & Energy Efficiency Toolkit (Stockholm, Ottawa data)
Resistance heater vs heat pump	Resistance heaters: ~50% range loss; Heat pumps: ~20% range loss	MDPI World Electric Vehicle Journal (2023)
Frequent stops / door cycling	Half of cold-weather energy increase from cabin heating with door opening	Gu et al., Cornell/TCAT (2025)
Idling / traffic delay	HVAC during idling is dominant energy contributor; idling exceeds driving time	LYMMO Orlando longitudinal analysis, MDPI (2025)

1.2 The latency problem

In a diesel operation, discovering a problem at 05:00 is inconvenient. In an electric operation, it is often too late. Because charging progress and energy feasibility must be managed hours in advance, the operation needs continuous visibility and early-warning signals.

Latency typically occurs because planning systems do not model chargers and grid limits, and charge management systems do not understand the duty plan. Without orchestration, humans attempt to bridge the gap manually, often in the most time-critical window.

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1.3 Two common cascade patterns

Overnight pull-out risk: a charger fault, cold weather, or delayed arrival means one or more vehicles miss their required State of Charge by departure time. The first duty fails and the recovery consumes the entire day.

Same-day opportunity charging cascade: a minor disruption at 14:00 shifts arrival times at a terminus charger. A missed charging window then propagates into an evening service failure. In these operations, the penalty for delay is non-linear.

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2 The Solution: Two-Layer Architecture

The fastest route to operational value is not rip-and-replace. Operators already own specialist tools for scheduling, charge management, and telematics. The challenge is the gap between them. A two-layer architecture closes that gap while preserving vendor independence.

2.1 Layer 1: Specialist foundation (keep what works)

Layer 1 consists of best-of-breed systems that remain in place: scheduling (blocks and duties), depot and opportunity charging control, and vehicle telematics. These systems continue doing what they do best.

2.2 Layer 2: The digital control tower (orchestration layer)

Layer 2 is the orchestration layer that continuously fuses; plan, live fleet state, and infrastructure constraints to create a single operational picture. It:

- Ingests planning data so actual-versus-planned can be evaluated at any moment.
- Normalises real-time State of Charge and location across OEMs and telematics providers.
- Models grid and charger constraints (power caps, curfews, priority rules) and their impact on feasibility.
- Detects conflicts early and identifies which routes are at risk.
- Proposes executable actions with operational guardrails and an audit trail.
- Maintains vendor independence through open interfaces so underlying systems can evolve without rebuilding the orchestration layer.

2.3 Multi-depot route complexity

As fleets scale, energy management becomes a network problem. Vehicles, chargers, and duties interact across depots. Orchestration enables cross-depot conflict resolution while respecting route and crew constraints.

2.4 Depot-specific grid constraints

Grid capacity and time-based restrictions differ by location. Orchestration makes those constraints explicit and optimises within them.

Depot	Max Capacity	Restriction
Central	2.4 MW	None
North	1.8 MW	50% cap 16:00–19:00
East	1.2 MW	No charging 07:00–19:00 (Nov–Mar)

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2.5 The standards that make it work

Open standards reduce integration cost, avoid lock-in, and increase resilience. A practical orchestration stack typically uses:

Standard	Function	Why It Matters
OCPP	Charger-to-system communication	Control any charger brand from one platform
OppCharge / EN 50696	Pantograph communication	Standardised interface for opportunity charging
VDV 463	Depot-to-charger integration	Tell the charger "95% by 06:00" and let it optimise
VDV 261	Vehicle preconditioning	Heat the bus on grid power, not battery
GTFS-RT	Real-time vehicle location	Feed live position into energy predictions

Five traits of an operations orchestration platform

Data-first
- A canonical operational model that makes constraints explicit and queryable

Extensible
- Adapters and open APIs/events so new systems and depots plug in without rework.

Continuous
- Deliver value in increments; keep integrating and improving without big-bang cutovers.

Orchestrated
- Workflow execution that turns decisions into actions across systems and people.

AI-enabling
- A clean, governed data backbone that allows optimisation and AI to be deployed safely

Orchestration delivers value proportional to the control it can exercise. Before committing to a deployment model, operators must assess what level of integration and command authority is achievable across their ecosystem.

2.6 Prerequisites and Operating Modes

The orchestration layer can operate at three maturity levels. Each delivers value, but the outcomes differ:

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Mode	Capability	Prerequisites	Typical Value
Visibility Only	Single operational picture; early warning alerts; manual intervention	Read-only telematics API; schedule data export; charger status feed	Decision latency reduction; baseline measurement; process discovery
Approve-Before-Execute	Ranked resolution proposals; human approval triggers execution	Read + limited write to CMS (charge priority); schedule amendment API or manual bridge	Guided workflows; reduced cognitive load; auditability; consistent decisions
Closed-Loop Optimisation	Auto-execute within guardrails; continuous optimisation; exception-only human involvement	Full bidirectional APIs to CMS, scheduling, telematics; contractual authority; validated data quality	15-25% energy savings; near-elimination of stranded incidents; operational headcount efficiency

Control Point Requirements

The following control points must be assessed during Phase 0. Where control is absent, identify the contractual or technical path to obtain it, or plan for a fallback operating mode:

System	Minimum (Visibility)	Target (Approve)	Optimal (Closed-Loop)
Charge Management (CMS)	Charger status feed; session history export	Priority adjustment API; target SoC by time (VDV 463)	Full OCPP pass-through or delegated scheduling authority
Scheduling System	GTFS export; daily block/duty feed	Real-time amendment notification; swap proposal acceptance	Bidirectional vehicle-block reallocation; crew constraint awareness
Telematics / OEM	SoC, position at 30-60 sec intervals	Sub-15 sec refresh; HVAC, auxiliary load visibility	Preconditioning commands (VDV 261); battery health indicators
Energy / Grid Interface	Tariff schedule import; capacity limit awareness	Real-time price signals; demand response event notification	OpenADR participation; V2G command capability

If Control Rights Are Missing

Where a third party (CPO, outsourced scheduling provider, OEM) controls a system:

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1. Document the gap in Phase 0 and quantify impact on achievable outcomes.
2. Negotiate API access or command delegation as part of contract renewal or variation.
3. Deploy at the achievable maturity level first; upgrade operating mode when control is obtained.
4. Consider whether a parallel manual workflow can bridge the gap until integration matures.

Orchestration at visibility-only level still delivers value (decision latency reduction, baseline measurement, process standardisation) but should not be seen as closed-loop optimisation.

2.7 What Orchestration Is Not

The orchestration layer occupies a specific position in the operational technology stack. Understanding what it does not do is as important as understanding what it does.

Orchestration Is Not...	Why The Distinction Matters	Relationship to Orchestration
A replacement for scheduling systems	Optibus, HASTUS, IVU.suite are sophisticated planning tools with deep domain functionality. Replacing them would require years and massive investment.	Orchestration reads from scheduling systems; it does not compete with their planning capabilities.
A replacement for charge management	Charge management systems control hardware, manage load balancing, and handle safety protocols. These are specialist functions.	Orchestration sends target outcomes (SoC by time); the CMS determines how to achieve them.
A telematics platform	Vehicle data collection requires OEM relationships, hardware integration, and telecommunications expertise.	Orchestration consumes telematics data; it does not collect it.
A BI / analytics tool	Business intelligence serves a different audience (executives, analysts) with different time horizons (historical, trend-based).	Orchestration operates in real-time; analytics can consume the data it generates.
A generic integration platform	iPaaS and ESB products move data but do not understand operational constraints or propose actions.	Orchestration includes integration but adds domain-specific decision support and workflow execution.

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The Unique Role of Orchestration

Orchestration is the cross-system operational decision layer that:

- Reduces decision latency by fusing data from multiple sources into a single operational picture
- Detects conflicts that span system boundaries (e.g., a scheduling decision that creates an energy problem)
- Proposes ranked, executable actions with operational guardrails
- Executes approved workflows across multiple systems and human processes
- Provides the governed data backbone that enables optimisation and AI to be deployed safely

Why Not Let Scheduling or Charging Vendors Expand?

Scheduling vendors (Optibus, IVU, etc.) and charging vendors (ChargePilot, Heliox, etc.) are expanding their capabilities. Why add another layer?

Vendor neutrality: Operators typically have multiple scheduling systems, multiple charger brands, and multiple OEMs. A vendor-specific solution creates lock-in and leaves gaps at system boundaries.

Execution focus: Scheduling vendors optimise plans; charging vendors optimise charge sessions. Neither is designed to execute real-time operational workflows that span both domains.

Incentive alignment: Specialist vendors are incentivised to expand their own footprint. An orchestration layer is incentivised to make all underlying systems work better together.

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3 Outcomes: What Orchestration Delivers

3.1 Outcome 1: Guarantee the morning pull-out

Pull-out reliability becomes a charging execution problem. Orchestration continuously simulates charging progress against the next departure wave and flags at-risk vehicles early enough to act (swap, reprioritise charging, or replan duties).

3.2 Outcome 2: Cut energy costs by 15-25%

Energy is one of the largest controllable levers in electric operations. Smart charging shifts consumption to lower-tariff windows, respects capacity limits, avoids peak-demand penalties, and uses preconditioning on grid power where possible.

3.3 Outcome 3: Decouple crew from chaos

Last-minute charge uncertainty drives last-minute crew changes, which drives disruption and cost. Orchestration stabilises assignments by providing verified charge feasibility and by proposing swaps that respect both vehicle and crew constraints.

3.4 Outcome 4: Absorb disruption without collapse

In electric fleets, small disruptions can create non-linear failures. Orchestration enables exception-based control: the operation focuses on a small number of true risks and follows guided workflows to resolve them before they cascade.

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4 Managing the Transition: The Human Element

Technology does not swap buses; people do. The most sophisticated orchestration algorithm is useless if the depot is unstaffed when a swap is required, or if the dispatcher lacks the authority to override a legacy roster.

Successful deployment is **20% software installation and 80% organizational transformation.** To bridge the gap between planning and reality, operators must evolve their human processes alongside their digital ones.

4.1 Mixed Fleets: Diesel and Electric as One Service

Most operators will run mixed diesel-electric fleets for 5-15 years during the transition. Orchestration should treat the passenger service as the unit of delivery, selecting the right asset for each duty based on capability, constraints, and availability, without creating parallel control rooms or operational silos.

Vehicle Capability Assessment

Every vehicle in the fleet should have a defined capability profile that the orchestration layer uses for allocation decisions:

Capability Attribute	Description	Allocation Impact
Propulsion type	Diesel, electric (depot-charged), electric (opportunity-charged), hybrid	Determines which routes are feasible and charging dependencies
Effective range (seasonal)	Validated range under current conditions (temperature, HVAC mode)	Determines duty feasibility without mid-route charging
Opportunity charging capability	Pantograph compatibility, fast-charge capability	Determines eligibility for routes requiring en-route charging
Emissions zone compliance	ULEZ, LEZ, zero-emission zone eligibility	Hard constraint for zone-restricted routes
Vehicle class	Standard, articulated, double-deck, accessibility features	Route and service type compatibility
Maintenance status	Days until scheduled maintenance, known defects	Affects availability and allocation priority

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Allocation Logic for Mixed Fleets

The orchestration layer applies a tiered allocation logic that respects both hard constraints and operator preferences:

1. Hard constraints (must be satisfied): Vehicle range covers duty distance with required buffer. Vehicle class matches route requirements. Emissions compliance for route zones. Driver qualified for vehicle type.

2. Preference constraints (optimised): Maximise electric miles where feasible. Minimise energy cost (prefer off-peak charging). Balance charger utilisation across depot. Minimise vehicle repositioning (same-lane assignments).

3. Fallback rules: If no electric vehicle can satisfy duty constraints, assign diesel. If multiple vehicles qualify, prefer the one with lowest opportunity cost (e.g., not needed for a later duty that requires higher SoC).

Interface to Rostering and Union Agreements

Mixed fleet orchestration must respect crew constraints that affect vehicle assignment:

- Driver qualifications: Not all drivers are certified for all vehicle types. The orchestration layer must maintain a driver-to-vehicle-class eligibility matrix.
- Union agreements: Some agreements specify minimum notice periods for vehicle type changes, or limits on electric vehicle assignments per driver per week (due to range anxiety concerns during transition).
- Duty continuity: Where possible, maintain driver-vehicle pairing within a shift to reduce cognitive load and enable drivers to report accurate vehicle-specific feedback.

These constraints should be codified in the orchestration layer as configurable rules, reviewed with union representatives, and updated as agreements evolve.

4.2 Vehicle capability assessment

Orchestration quality depends on vehicle truth. A Phase 0 assessment should validate: telematics accuracy, battery health indicators, charge curves, preconditioning support, and route energy profiles under seasonal extremes.

4.3 Organisational readiness and change management

Technology without organisational change delivers dashboards, not outcomes. Orchestration only works when decision rights, staffing, and yard processes can execute a recommended action in minutes, not morning meetings.

Before deploying software, assess whether the operation can act on the intelligence it will receive. The scorecard below converts 'change management' into operational readiness criteria.

The Orchestration Readiness Scorecard

"Orchestration Ready" depots exhibit specific characteristics that differ from legacy diesel operations. Use this scorecard to assess your operational agility *before* deploying software.

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Factor	Legacy / Diesel Operation	Orchestration Ready	Score (0-2)
Decision authority	Hierarchical: Overnight supervisors must escalate roster changes to a manager (often next morning).	Delegated: Overnight staff have full authority to execute system-recommended swaps to protect the pull-out.	
Operational silos	Fragmented: Dispatch owns "On-Time Performance"; Facilities owns "Energy Bill." Teams rarely collaborate.	Unified: Shared KPIs across dispatch, depot, engineering, and energy; conflicts resolved through defined rules.	
Physical yard agility	Static: Buses park and stay; shunting is reactive; charger bay occupancy not actively managed.	Dynamic: 24/7 ability to move vehicles between lanes/chargers via shunters or standard yard protocols.	
Dispatcher role	Firefighter: Reacts to breakdowns. Relies on "gut feel" and experience.	Controller: Monitors "exception-based" alerts. Trusts data-driven predictions to solve problems before they occur.	
Union engagement	Skeptical: Range monitoring viewed as surveillance or a safety risk.	Collaborative: Drivers trust the "Verified Range" display. Unions are partners in defining fair assignment rules.	

Scoring guide: 0 = not in place, 1 = partially in place, 2 = institutionalised. Total score indicates readiness for automation vs. visibility-only deployment.

4.4 Operating model and governance

Orchestration works when decision rights and platform guardrails are explicit. The recommended operating model is federated: a small central platform function sets standards and runs the backbone, while depots and control rooms own the day-to-day operational playbooks and escalation protocols.

Platform owner (Ops/Engineering): accountable for outcomes (pull-out reliability, stranded incidents, energy cost).

Platform team: canonical data model, integration patterns, security, monitoring, and change control for interfaces and workflows.

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Depot/control leadership: owns operating procedures, exception handling, and the authority to execute swaps, charge-plan overrides and roster changes in minutes.

Data governance: clear ownership of key data products (vehicle state, charger status, duty plan) and agreed data quality SLAs.

Vendor onboarding: contractual requirements for APIs/events, test environments and backward-compatible interface versioning.

4.5 Redefining the dispatcher: from firefighter to air traffic controller

In a manual environment, a dispatcher's value is defined by how fast they can fix a broken plan. In an orchestrated environment, their value is defined by how well they manage the network to prevent breaks. This requires a fundamental shift in mindset.

Trust the "Silent" Dashboard:

Dispatchers must learn that no news is good news. If the system isn't alerting, the plan is working. The instinct to constantly check and double-check is natural, but it undermines the efficiency gains orchestration provides.

The Human-in-the-Loop Protocol:

A practical protocol should define three categories of actions:

- **Auto-execute (pre-approved):** low-risk actions such as charger reprioritisation within defined caps.
- **Approve-before-execute:** actions that affect service delivery, such as cross-depot swaps or duty reassignments.
- **Forbidden or escalate:** actions with safety or contractual implications.

The technology suggests the mathematically best option; the human ensures it is the operationally feasible one. The dispatcher knows that Bus 402 has a broken wing mirror—the algorithm does not.

4.6 Bridging the driver trust gap

Drivers are the ultimate end-users of electrification. If they do not trust the vehicle's range, they will return to the depot with 20% charge "just to be safe," destroying the efficiency gains you planned for.

Transparency:

Give drivers access to the same "Verified SoC" data the control room sees via mobile apps or depot screens. When drivers and dispatchers see the same numbers, trust builds naturally.

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Feedback loops:

Create a formal channel for drivers to report when reality disagrees with prediction (e.g., "System said 15% remaining, but the dashboard flashed red"). Treat these reports as model improvement data, not blame.

Coaching, not punishment:

If efficiency feedback is used, frame it as coaching rather than surveillance. "Smart Driving" features can reward efficient behaviour, but must be carefully positioned to maintain morale during the transition. Align data use with union agreements before deployment.

4.7 Data sovereignty: owning your intelligence

Orchestration increases the value of operational data. Operators should retain ownership of the canonical data model, the event history, and the optimisation logic. This enables vendor independence and ensures that insights generated from the operation remain an operator asset.

The bottom line: You cannot overlay a dynamic digital platform onto a static, rigid organizational structure. If you buy the software but keep the silos, you will simply have a more expensive way to fail. Use Phase 0 to redesign your workflows, not just your IT network.

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5 The Financial Case

5.1 Investment profile

A typical programme follows three phases. Investments vary by fleet size, system maturity, and integration complexity, but the pattern is consistent: integrate first, optimise second, then scale.

Phase	Timeline	Investment & Focus
Foundation	Months 0–12	€1.5–2.5M: Core integration, data hub, operational visibility
Pilot Corridor	Months 12–24	€2.0–3.5M: Smart charging, predictive alerts, crew-charge sync
Fleet-Wide Scale	Months 24–36	€1.5–2.5M: Network rollout, advanced AI, V2G readiness

5.2 Return profile

Benefits compound as the platform moves from visibility to closed-loop optimisation. The table below shows typical benefit categories.

Benefit Category	Year 1	Year 2	Year 3
Energy cost reduction	€0.5–1M	€1–2M	€2–3M
Avoided penalties & compensation	€0.3–0.5M	€0.8–1.2M	€1.5–2M
Reduced manual coordination	€0.2–0.4M	€0.5–0.8M	€0.8–1.2M
Fleet size optimisation	—	€0.5–1M	€1–2M

5.3 Beyond cost reduction: strategic value

- **Resilience:** fewer service failures, faster recovery, and a measurable reduction in stranded-vehicle incidents.
- **Scalability:** a repeatable pattern to add depots, chargers, routes, and OEMs without redesigning the control room.
- **Procurement strength:** evidence-based reliability and efficiency improvements can strengthen PSO tenders and stakeholder confidence.
- **V2G readiness:** bidirectional charging and grid services are market-dependent, but orchestration is the prerequisite layer for safe participation.

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Value stack for CFOs: foundation, extension and option value

Foundation value: single operational picture, decision latency reduction, fewer stranded incidents, measurable pull-out reliability.

Extension value: charging and resource optimisation (energy, depot capacity), pilot corridor orchestration, predictive maintenance decision support.

Option value: future use-cases (V2G, new depots/OEMs, new tariffs and grid constraints) delivered on the same backbone, avoiding re-integration.

5.4 Assumptions Behind the Financial Case

The investment and return profiles in sections 5.1-5.2 are based on the following reference scenario. Actual results vary by fleet size, tariff structure, baseline maturity, and operating model.

Parameter	Reference Scenario
Fleet size	200-300 electric buses across 2-3 depots
Annual energy consumption	15-20 GWh (approximately 75,000 kWh per bus per year)
Tariff structure	Time-of-use with 2-3x peak/off-peak differential
Baseline peak charging exposure	35-45% of energy consumed at peak rates
Baseline pull-out reliability	94-96%
Stranded incidents (baseline)	8-15 per year across fleet
Manual coordination FTE	2-4 FTE equivalent spent on charging/dispatch coordination
PSO penalty regime	€500-2,000 per missed departure; €5,000-15,000 per stranded incident

5.5 Sensitivity by Fleet Size

Benefits scale with fleet size and energy consumption. Smaller fleets achieve lower absolute savings but often faster payback due to simpler integration. Larger fleets unlock network effects (cross-depot optimisation) but require greater change management investment.

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Fleet Band	Year 1 Investment	Annual Benefit (Yr 3)	€ per Bus/Year	Payback
50-100 buses	€0.8-1.2M	€1.5-2.5M	€15,000-25,000	18-24 months
100-250 buses	€1.2-2.0M	€3.0-5.0M	€12,000-20,000	12-18 months
250-500 buses	€1.8-3.0M	€5.0-9.0M	€10,000-18,000	12-15 months
500+ buses	€2.5-4.0M	€8.0-15.0M	€8,000-15,000	10-14 months

Key Sensitivity Factors

The following factors have the largest impact on achievable savings:

- Tariff volatility: Operators with high peak/off-peak differentials (3x+) or exposure to spot market pricing see 20-30% higher energy savings than those on flat tariffs.
- Baseline charging efficiency: Fleets already running some form of smart charging achieve 10-15% lower incremental savings but benefit more from reliability and coordination improvements.
- Grid constraints: Depots with capacity limits or time-based restrictions unlock additional value through constraint-aware scheduling that would otherwise require costly infrastructure upgrades.
- Opportunity charging dependency: Operations relying on en-route charging see higher value from trajectory management and cascade prevention than depot-overnight models.

What Must Be True for 15-25% Energy Savings

- Tariff structure with meaningful peak/off-peak differential (minimum 1.5x, ideally 2x+)
- Controllable charging windows (ability to shift at least 40% of energy consumption)
- Command authority over charge management (approve-before-execute minimum)
- Charger utilisation headroom (not already at 95%+ overnight capacity)
- Accurate SoC data (validated telematics with <5% error margin)

Where these conditions are partially met, scale expectations accordingly. Phase 0 should quantify each factor for the specific operation.

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6 Implementation Roadmap

A Phase 0 assessment (typically 4-6 weeks) establishes baseline metrics, validates integration points, and produces a quantified business case. It is the fastest way to reduce risk before committing to a multi-year programme.

6.1 Phase 0: Readiness and baseline (4-6 weeks)

- Map current system landscape and integration points (scheduling, charging, telematics, energy tariffs).
- Define the canonical data model and minimum viable control tower views.
- Measure baseline KPIs (pull-out, energy cost structure, stranded incidents, manual coordination effort).
- Assess organisational readiness using the scorecard and agree the human-in-the-loop operating protocol.

Phase 0 is commonly sized at €50-100K and is designed to de-risk the subsequent phases by turning assumptions into measured facts.

6.2 Phased delivery

The three delivery phases below align technical build-out with operational maturity gates.

Phase 1: Foundation and visibility (Months 0-12)

- **Goal:** establish the single operational picture.
- **Deliverables:** integrations across scheduling, charging (depot and opportunity), and telematics; real-time State of Charge view; basic alerting.
- **Benefit gate:** dispatchers reclaim time spent gathering information; at-risk pull-outs move from 'morning of' to 'hours before'.

Phase 2: Pilot corridor orchestration (Months 12-24)

- **Goal:** prove value on a contained part of operations (one depot, corridor, or route family).
- **Deliverables:** smart charging optimisation, predictive pull-out alerts, guided resolution workflows, and agreed operating protocol.
- **Benefit gate:** measurable energy cost reduction and fewer stranded incidents in the pilot scope.

Phase 3: Fleet-wide scale (Months 24-36)

- **Goal:** scale proven capabilities across the network.
- **Deliverables:** multi-depot conflict resolution, advanced optimisation, rollout of driver-facing transparency features, and V2G pilots where relevant.
- **Benefit gate:** platform becomes self-funding through recurring savings and reliability improvements.

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7 Next Steps

7.1 Key performance indicators for orchestrated operations

KPI	What It Measures	Why It Matters
Pull-out reliability rate	% of scheduled departures achieved on time	Service delivery, SLA compliance
Peak-rate charging %	Share of energy consumed at peak tariff rates	Direct cost lever, often 20–30% savings opportunity
Stranded vehicle incidents	Vehicles unable to complete duty due to energy	Service disruption, recovery cost
SoC deviation frequency	How often actual SoC falls below planned trajectory	Leading indicator of cascade risk
Charging plan adherence	% of planned charging events executed as scheduled	Opportunity charging reliability

7.2 Self-assessment questionnaire

3 Questions to answer

What is your current pull-out reliability rate? If you do not measure it, start there.

How much are you spending on energy, and what share is exposed to peak tariffs or demand charges?

When a driver calls in sick at 05:00, how long until dispatch knows which charged vehicles can cover without creating an energy failure later in the day?

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Leadership checklist: platform readiness in 30 minutes	
• Is there an agreed case for change and a quantified baseline (pull-out reliability, stranded incidents, energy costs, dispatcher workload)?	
• Who is the single accountable owner for outcomes and adoption across engineering, scheduling and operations?	
• Do we have funding for Phase 0/1 and an agreed path to scale (not just a pilot budget)?	
• What is the first executable use case we will industrialise (not just visualise), and what are the exit criteria?	
• Are supplier contracts aligned to open APIs/events, data ownership and interface change control (so new tools plug in, rather than creating new silos)?	
• How will success be governed and measured month-by-month (KPIs, decision latency, adoption, reuse of components)?	

7.3 Getting started

Start with Phase 0. Map systems, quantify baseline metrics, validate constraints, and agree the operating protocol and organisational changes required to act on orchestration recommendations.

The bus that does not charge does not run. The fleet that does not orchestrate cannot scale electrification with confidence.

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8 Appendix A: Technical Architecture

This appendix summarises a reference architecture for an orchestrated depot digital control tower. It is intentionally vendor-neutral and aligns with open standards.

Layer	Purpose	Typical components	Interfaces / standards
Layer 1 - External systems	Systems of record and specialist controllers	Scheduling (Optibus, HASTUS, IVU); Charge management; Telematics; Grid tariffs	REST APIs; GTFS/GTFS-RT; OCPP; VDV 463; VDV 261
Layer 2 - Integration	Normalise data and create single operational picture	Adapter framework; canonical data model; event streaming; time-series store	OpenAPI; OAuth 2.0; Webhooks; WebSocket
Layer 3 - Orchestration	Detect conflicts, run feasibility checks, propose actions	State estimator; constraint manager; optimisation solver; command dispatch	Policy/rules engine; MILP optimisation
Layer 4 - Intelligence	Forecast energy and operational risk	Energy consumption model; SoC trajectory forecasting; anomaly detection	Model serving APIs; feature store
Layer 5 - Interfaces	Role-based workflows	Control tower dashboard; driver mobile app; management analytics; external API	React/Web; Push notifications; BI connectors; OpenAPI

8.1 Architecture principles

Open standards: REST APIs, modular adapters, and cloud-native deployment options.

Modular integration: add systems incrementally; avoid single points of failure in integration design.

Vendor independence: platform-agnostic design to prevent lock-in and keep procurement flexible.

Security by design: encrypted communications, segmented networks, and data sovereignty controls.

8.2 Vendor onboarding requirements

To keep the control tower vendor-neutral, require every system supplier to meet a minimum interface contract:

• Documented REST APIs and/or event streams for operational state changes (vehicle, charger, duty, depot constraints).

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• Data portability: ability to export raw and derived data products, with stable identifiers and schemas.

• Test harness and sandbox environment (or simulator) to validate integrations and regression-test workflows.

• Backward-compatible versioning and change notice for interface updates; no breaking changes without agreed windows.

• Security and access controls aligned to operator policies (OAuth2/OpenID where applicable, least-privilege roles).

8.3 Accountability, Safety, and Cybersecurity

Orchestration introduces automated decision support and, in some cases, automated execution. This creates new questions about accountability, audit requirements, and operational safety.

Decision Accountability

Clear accountability must be established for each decision category:

Decision Type	Accountable Party	Audit Requirement
Auto-executed (pre-approved)	Operator (via policy approval)	Full logging of inputs, constraints, and action taken
Human-approved	Approving dispatcher/supervisor	Decision logged with approver ID, timestamp, and context presented
System recommendation rejected	Rejecting human	Rejection reason logged; alternative action recorded
Algorithm design	Platform provider	Model documentation, validation results, and change history

Explainability and Audit Trail

Every recommendation and action must be explainable. The audit trail should capture:

- Inputs: What data was available at decision time (vehicle state, constraints, forecasts)
- Constraints: What rules and limits were applied
- Alternatives: What options were considered and why the recommended action ranked highest
- Outcome: What action was taken and what the result was
- Deviations: Cases where actual outcome differed from prediction (for model improvement)

Audit data should be retained for a minimum of 12 months and be exportable for regulatory or contractual review.

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Cybersecurity and OT Boundaries

Charge infrastructure is increasingly treated as Operational Technology (OT) with heightened security requirements. Orchestration platforms that issue commands to charging systems must address:

- Network segmentation: Clear boundaries between IT (orchestration platform) and OT (charger control) networks, with controlled crossing points.
- Authentication and authorisation: All commands must originate from authenticated sources with appropriate role-based permissions.
- Command validation: Safety-critical commands (e.g., immediate charge stop) should have secondary validation or rate limiting.
- Incident response: Defined procedures for suspected compromise, including ability to isolate the orchestration layer without halting charging operations.

Security architecture should be reviewed with the operator's IT security team and, where applicable, aligned to frameworks such as IEC 62443 for industrial automation security.

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9 Appendix B: Proof Point - AIRHART (Airport Operations Orchestration Platform)

AIRHART is a live example of the same platform pattern described in this paper: an operations orchestration layer that unifies data, coordinates workflows and enables continuous optimisation in a time-critical environment. This appendix is included as a proof point and as a transferable blueprint; the main paper remains vendor-neutral and depot-focused.

9.1 The platform pattern in one page

Data-first: A canonical operational model that makes constraints explicit and queryable.

Extensible: Adapters and open APIs/events so new systems and depots plug in without rework.

Continuous: Deliver value in increments; keep integrating and improving without big-bang cutovers.

Orchestrated: Workflow execution that turns decisions into actions across systems and people.

AI-enabling: A clean, governed data backbone that allows optimisation and AI to be deployed safely.

9.2 Delivery route map (proof of value to scaling)

Stage	Primary goal	Typical deliverables	Depot transfer (what it enables)
Proof of value	Demonstrate an executable workflow on a limited scope using real integrations.	Canonical model slice; 2-3 adapters; alerting and decision dashboard; one closed-loop workflow with human approval; baseline KPIs.	Phase 1 foundation: visibility plus a first 'approve-before-execute' workflow (e.g., charger reprioritisation tied to duty risk).
Industrialisation	Harden the backbone and make the workflow repeatable across assets, shifts and exceptions.	Monitoring and resilience; role-based access; interface versioning; operating protocol; training and adoption plan; playbooks; expanded data products.	Phase 1-2: stable control tower, repeatable depot operating protocol, reduced decision latency.
Scaling	Roll out to additional sites/use cases without re-integrating the world.	Reusable adapter library; configuration-driven rules; governance model; continuous delivery cadence; portfolio of optimisation/AI use cases.	Phase 3: multi-depot scale, new OEMs/chargers/routes onboarded through standard contracts.

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9.3 Blueprint mapping (airport to depot)

AIRHART capability (pattern)	Depot analogue	Implementation hint (ties to Appendix A)
Single operational picture	Single source of truth across planning, charging, telematics and depot constraints	Canonical operational model; event ingestion; data products for duty risk and charger/vehicle state.
Workflow orchestration	Executable actions across systems and people (auto / approve / escalate)	Workflow engine; role-based approvals; audit trail; playbooks and escalation protocols.
Constraint awareness	Timetable, energy, capability, depot capacity and grid constraints treated as one system	Rules/constraints service; scenario evaluation; conflict detection and recommendations.
Continuous improvement	Incremental onboarding of depots, routes, OEMs, chargers, tariffs	Adapter library; configuration over custom code; contract-first APIs and versioning.
AI and optimisation	Predictive and prescriptive support that is trusted and governable	MLOps hooks; model monitoring; human-in-the-loop; bias and drift checks on operational recommendations.

9.4 Transfer kit (what to reuse)

• Canonical model patterns: how to represent operational state, constraints and events in a stable schema.

• Adapter patterns: how to integrate legacy tools safely (polling vs events, id mapping, retry and resilience).

• Operating protocol templates: auto/approve/escalate decision rights, audit trail and exception playbooks.

• Governance blueprint: platform ownership, interface change control, data product SLAs and vendor onboarding rules.

• Delivery cadence: release train, regression testing for workflows, and KPI gates for scaling.

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10 Appendix C: Interface Reality by Fleet Archetype

The standards referenced in Section 2.5 are not universally adopted. This appendix maps realistic interface availability by region and charging topology to help operators assess integration complexity.

C.1 Regional Standards Adoption

Standard	DACH	Nordics	UK	Benelux	France
OCPP 1.6/2.0	High	High	High	High	High
VDV 463	High	Medium	Low	Medium	Low
VDV 261	High	Low	Low	Low	Low
OppCharge/EN 50696	High	High	Medium	High	Medium
GTFS-RT	High	High	High	High	High
ITxPT	Medium	High	Low	Medium	Medium

High = widely deployed, standard interface available. Medium = partial adoption, may require adapter or negotiation. Low = rare, typically requires custom integration or alternative approach.

C.2 Typical Vendor Stacks by Charging Topology

Depot-overnight (AC/DC):

Common CMS vendors: ChargePilot (The Mobility House), Heliox, ABB, Siemens. Most support OCPP 1.6/2.0. VDV 463 support varies; require explicit confirmation during procurement.

Opportunity charging (Pantograph):

Common vendors: Schunk, Siemens, ABB, Heliox. OppCharge/EN 50696 generally supported for interoperability. Integration with depot CMS varies; some operators run separate systems.

Mixed topology:

Requires unified orchestration layer to bridge depot and opportunity charging. Some CMS vendors offer both; others require integration between systems. This is where orchestration adds the most value.

C.3 Where Adapters Are Required

Expect custom adapter development for:

- OEM telematics platforms (each OEM has proprietary APIs; GTFS-RT conversion typically required)

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- Legacy scheduling systems without REST APIs (file-based integration or database access)
- Proprietary charge management systems (especially older installations)
- Municipal or regional energy management interfaces (varies by DSO/TSO)

Phase 0 should include a detailed interface audit for each system in scope, documenting API availability, data freshness, and any contractual restrictions on data access.

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